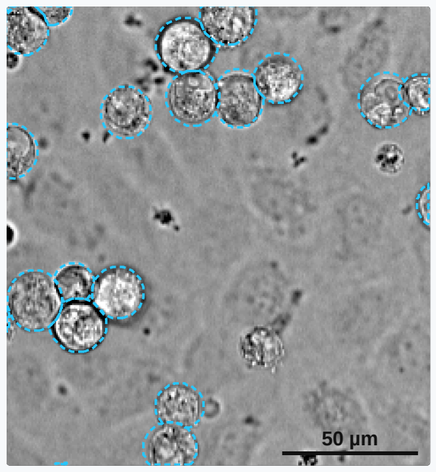
## a When a pre-trained model performs poorly, find the mismatch first

field 1 · field 2 (F1 per field) · model · manual annotation

### 1 Check the normalisation

tile: forgot to normalise

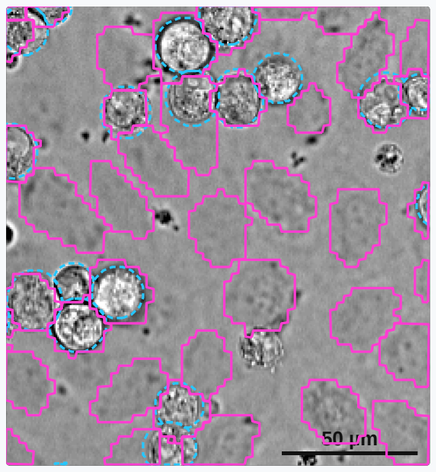



F1 (mean of 2 fields)

| | |
|---|---|
| **forgot to normalise** | **0.00** |
| percentile (default) | 0.59 |
| min–max | 0.61 |
| histogram equalisation | 0.51 |
| CLAHE | 0.51 |

### 2 Check the object scale

tile: input downsampled 4×

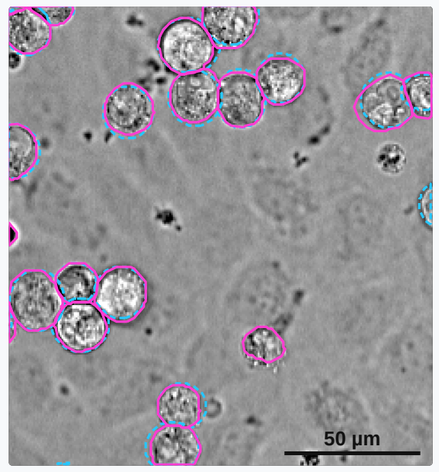



F1 (mean of 2 fields)

| | |
|---|---|
| **downsampled 4×** | **0.30** |
| downsampled 2× | 0.41 |
| native (default) | 0.59 |
| upsampled 2× | 0.56 |

### 3 Check the post-processing

tile: low-contrast objects removed




F1 (mean of 2 fields)

| | |
|---|---|
| none (default) | 0.59 |
| stricter probability threshold | 0.59 |
| looser flow threshold | 0.56 |
| **low-contrast objects removed** | **0.86** |

**4** If nothing works, fine-tune or train

## b Predicted images: what will you measure?

**Used as an intermediate step**

visualise · segment · detect · register

→ **Validate the downstream output**

e.g. segment on the predicted image, compare with annotations of acquired images

**Used as the measurement itself**

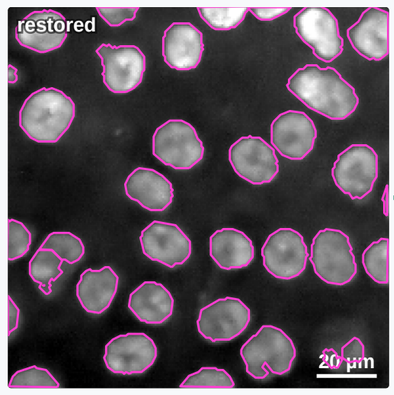
intensity · abundance · localisation

→ **Validate the measurement against acquired data**

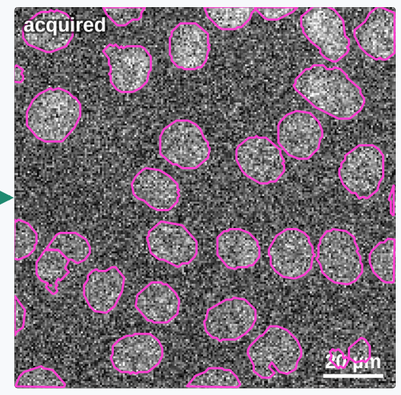
e.g. predicted vs acquired fluorescence, in independent paired samples

**Label predicted images as predicted; do not describe predicted fluorescence as measured.**

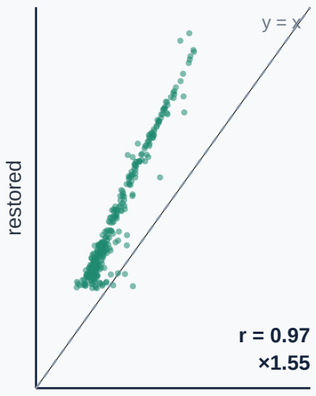
## c Denoising: measure intensity in the acquired image




mask from here

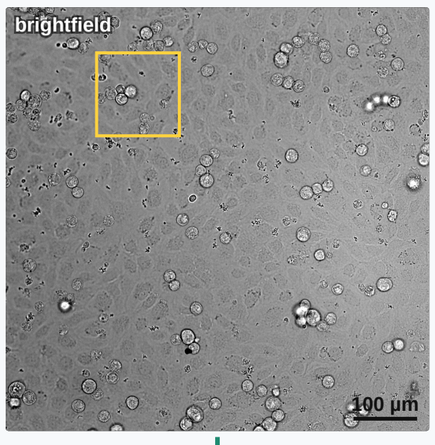



measure here

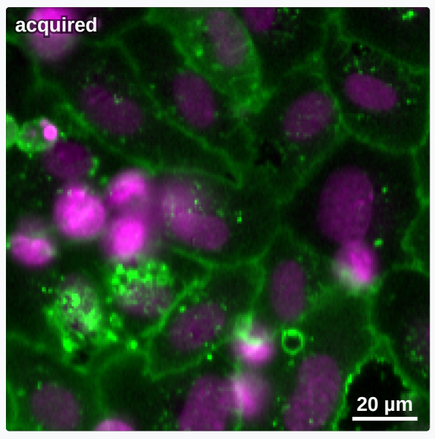



## d Artificial labelling: validate what you will measure

model · manual, annotator 1 · manual, annotator 2



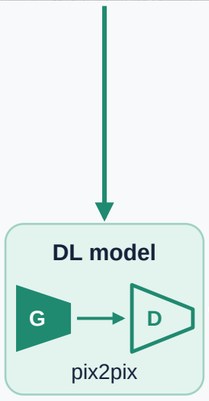

DAPI + PECAM-1



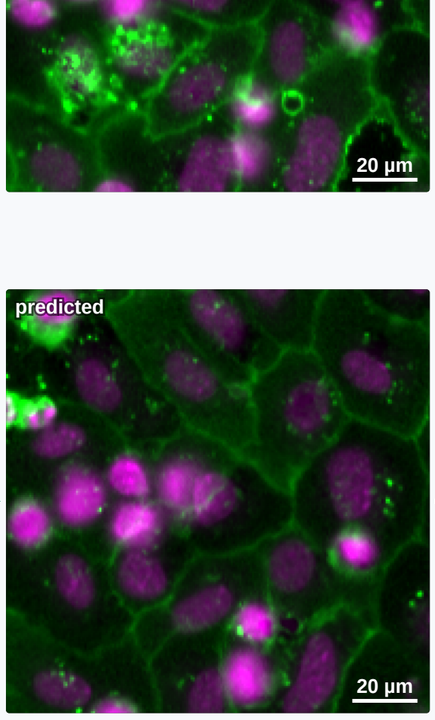

manual annotation

model segmentation

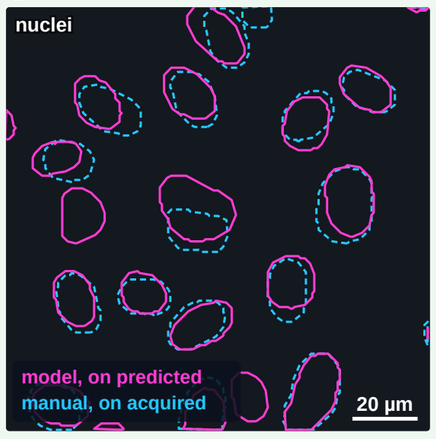
DL model




pix2pix

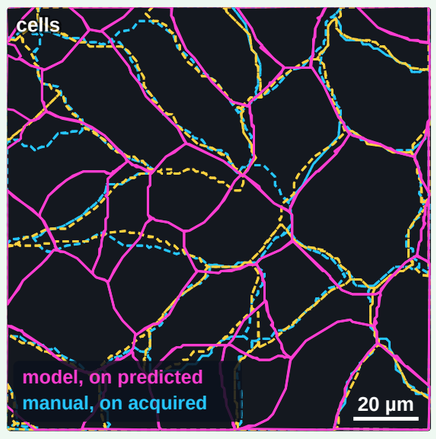



**Intermediate step: to aid segmentation**




**Nuclei · F1**

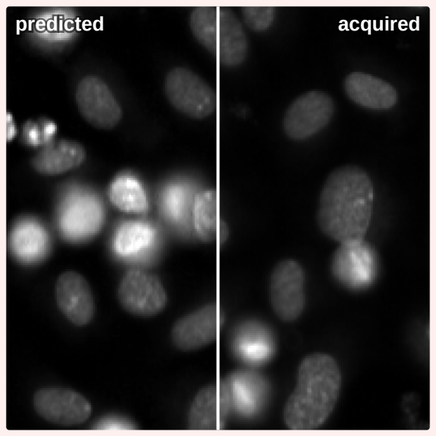
| | |
|---|---|
| model vs manual nuclei | 0.83 |



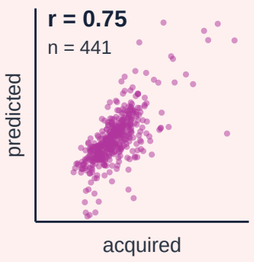

**Cells · F1**

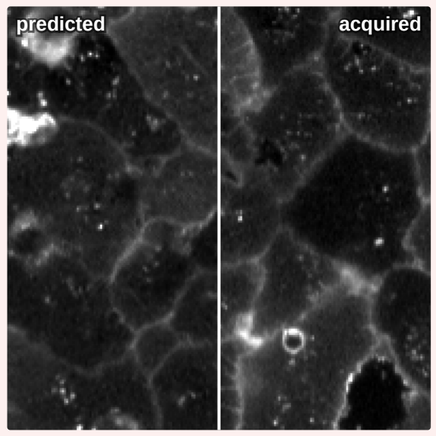
| | |
|---|---|
| annotator 1 vs 2 | 0.90 |
| model vs annotator 1 | 0.78 |
| model vs annotator 2 | 0.75 |

**Measurement itself: intensity**

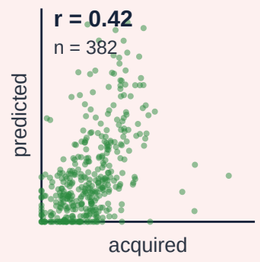



**DAPI, per nucleus**



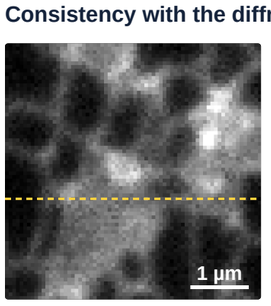




**PECAM-1, per cell**

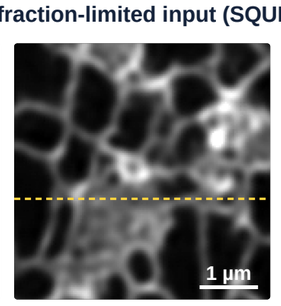



## e Super-resolution: check against the input and, where possible, higher-resolution data

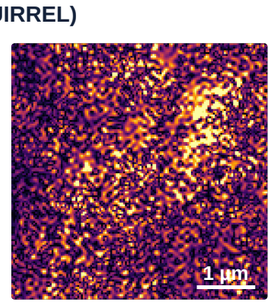
**Consistency with the diffraction-limited input (SQUIRREL)**




low-res input




predicted




error map

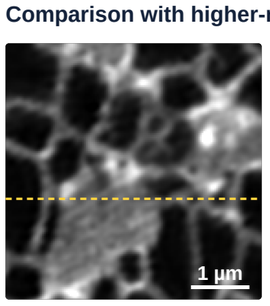
0 · error (a.u.) · high

**RSP with the input**

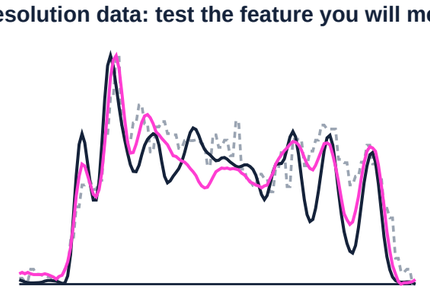
| | |
|---|---|
| predicted | 0.98 |
| higher-res | 0.93 |

0.90 · 0.95 · 1.00

**Comparison with higher-resolution data: test the feature you will measure**




higher-res




**Tubule width (FWHM, nm), 1,301 locations**

| | |
|---|---|
| low-res input | 293 |
| predicted | 199 |
| higher-res | 212 |

100 · 200 · 300 · 400

**predicted − higher−res, same location: −9 nm (IQR −28 to +7)**